

PR-PL: A Novel Prototypical Representation Based Pairwise Learning Framework for Emotion Recognition Using EEG Signals

Rushuang Zhou¹, Zhiguo Zhang¹, *Member, IEEE*, Hong Fu², Li Zhang³, Linling Li³, Gan Huang³, Fali Li⁴, Xin Yang³, Yining Dong³, Yuan-Ting Zhang³, *Fellow, IEEE*, and Zhen Liang³, *Member, IEEE*

Abstract—Affective brain-computer interface based on electroencephalography (EEG) is an important branch in the field of affective computing. However, the individual differences in EEG

emotional data and the noisy labeling problem in the subjective feedback seriously limit the effectiveness and generalizability of existing models. To tackle these two critical issues, we propose a novel transfer learning framework with Prototypical Representation based Pairwise Learning (PR-PL). The discriminative and generalized EEG features are learned for emotion revealing across individuals and the emotion recognition task is formulated as pairwise learning for improving the model tolerance to the noisy labels. More specifically, a prototypical learning is developed to encode the inherent emotion-related semantic structure of EEG data and align the individuals' EEG features to a shared common feature space under consideration of the feature separability of both source and target domains. Based on the aligned feature representations, pairwise learning with an adaptive pseudo labeling method is introduced to encode the proximity relationships among samples and alleviate the label noises effect on modeling. Extensive results on two benchmark databases (SEED and SEED-IV) under four different cross-validation evaluation protocols validate the model reliability and stability across subjects and sessions. Compared to the literature, the average enhancement of emotion recognition across four different evaluation protocols is 2.04% (SEED) and 2.58% (SEED-IV).

Index Terms—EEG, emotion recognition, affective brain-computer interface, transfer learning, affective computing.

I. INTRODUCTION

AFFECTIVE computing is a fast growing interdisciplinary research field and is attracting researchers' attention from different areas including computer science, neuroscience, psychology, and signal processing [1]. Recently, electroencephalography (EEG) based emotion recognition has become an increasingly important topic for affective computing and human sentiment analysis [2], [3]. A proper design of EEG-based emotion recognition models is helpful for facilitating data processing, benefiting discriminant feature characterization, and lightening model performance. Currently, there exist two main critical issues in EEG-based emotion recognition models. One is *individual differences*: how to build a generalized affective computing model which could adapt to the remarkable individual differences and detect discriminative features in the simultaneously collected EEG signals; and another is *noisy labeling*: how to build a reliable and stable affective computing model which is more tolerance to the contaminated label information.

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Rushuang Zhou is with the School of Biomedical Engineering, Medical School, Guangdong Provincial Key Laboratory of Biomedical Measurements and Ultrasound Imaging, Shenzhen University, Shenzhen 518060, China, and also with the Department of Biomedical Engineering, Hong Kong Center for Cerebro-Cardiovascular Health Engineering (COCHE), Hong Kong Science and Technology Park, City University of Hong Kong, Kowloon Tong, Hong Kong (e-mail: rrushuang2-c@my.cityu.edu.hk).

Zhiguo Zhang is with the Institute of Computing and Intelligence, Harbin Institute of Technology, Shenzhen 150001, China, and also with Peng Cheng Laboratory, Shenzhen 518066, China (e-mail: zhiguo Zhang@hit.edu.cn).

Hong Fu is with the Department of Mathematics and Information Technology, The Education University of Hong Kong, Ting Kok, Hong Kong (e-mail: hf@eduhk.hk).

Li Zhang, Linling Li, Gan Huang, Xin Yang, and Zhen Liang are with the School of Biomedical Engineering, Medical School, Guangdong Provincial Key Laboratory of Biomedical Measurements and Ultrasound Imaging, Shenzhen University, Shenzhen 518060, China (e-mail: lzhang@szu.edu.cn; lilinling@szu.edu.cn; huanggan@szu.edu.cn; yangxinknow@gmail.com; zhenliang.szu@gmail.com).

Fali Li is with the The Clinical Hospital of Chengdu Brain Science Institute, MOE Key Lab for Neuroinformation, School of Life Science and Technology, Center for Information in Medicine, University of Electronic Science and Technology of China, Chengdu 610056, China (e-mail: fali.li@uestc.edu.cn).

Yining Dong is with the School of Data Science, Hong Kong Center for Cerebro-Cardiovascular Health Engineering (COCHE), Hong Kong Science and Technology Park, City University of Hong Kong, Kowloon Tong, Hong Kong (e-mail: yinidong@cityu.edu.hk).

Yuan-Ting Zhang is with the Department of Biomedical Engineering, Hong Kong Center for Cerebro-Cardiovascular Health Engineering (COCHE), Hong Kong Science and Technology Park, City University of Hong Kong, Kowloon Tong, Hong Kong (e-mail: yt.zhang@cityu.edu.hk).

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Recently, more and more researchers have focused on applying transfer learning methods to alleviate the individual differences in EEG signals [4], [5], [6], [7], [8], [9] and improve feature invariant representation [10], [11], [12]. Under consideration of the feature distribution of both the source domain (labeled data with known distribution) and the target domain (unlabeled data with unknown distribution), the transfer learning methods try to minimize the distribution difference between the source domain and the target domain by approximately satisfying the assumption of independent and identical distribution and can consequently realize a higher recognition performance on the target domain. Through a domain-shifting strategy, the invariant feature representations across different domains are learned and the relationships between the learned features and emotional labels are explored. Currently, most existing EEG-based emotion recognition models are based on the transfer learning framework, in which both the source domain (data and label) and the target domain (data only) are seen during model training. For example, Li et al. [5] proposed a multisource transfer learning method with two different stages. In the first stage, appropriate samples were selected from the existing source domain. In the second stage, a style transfer mapping was implemented to alleviate the differences between the selected source samples and the unknown target samples. The results showed the proposed transfer method outperformed the non-transfer method with an improvement of 12.72% on the public SEED database [13] with a three-class classification problem (negative, neutral, and positive). Inspired by neuroscience findings that different emotions would lead to different brain reactions, Li et al. [6] proposed a novel R2G-STNN network to integrate the EEG spatial-temporal dynamics at the local and global brain areas and realize an efficient emotion recognition performance together with a domain shift learning. More details about current EEG-based emotion recognition models with the transfer learning algorithms are presented in Section II.

For video-evoking EEG-emotion experiments, subjects may not always be able to accurately react to the intended emotions, and at the same time may not be able to accurately describe and feedback on their emotional changes. This would bring label noise to the emotional information annotation of EEG samples and further lead to a negative impact on the model performance [14]. To tackle this issue, Zhong et al. [8] developed an emotion-aware distribution learning method (RGNN), in which they blurred the label information by changing the one-hot label representation $(1, 0, 0)$ to $(1 - \frac{2\epsilon}{3}, \frac{2\epsilon}{3}, 0)$ and trained the model to be less sensitive to the label noise. However, the model performance would greatly rely on the selection of ϵ value, and an optimal ϵ value selection could be different for different databases and different individuals. On the other hand, current EEG-based emotion recognition models are mainly realized by *pointwise learning*, which trains a model to be close to the given labels ($y = f(x)$). In other words, if the given labels are fully contaminated by noises, the model learning process could be easily affected. Contrarily, *pairwise learning* is to train a model to learn the relative associations between pairs of instances ($f(x_i, x_j)$) and encode the proximity among samples with less reliance on labeling. Thus, pairwise

learning has achieved tremendous success in tackling the noisy labeling issue in a number of real-world applications [15], [16], [17], [18].

In this paper, we formulate the EEG-based emotion recognition tasks as a pairwise learning problem and propose a novel transfer learning framework with prototypical representation based pairwise learning (which is termed as *PR-PL* below). Here, we model the relative relationship between pairs of EEG samples in terms of prototypical representations, which is advantageous to pointwise learning especially when the high-precision labels are difficult to obtain [19]. The major novelties of the proposed PR-PL are summarized as follows. (1) We propose a novel transfer learning framework to tackle the individual differences and noisy labeling issues which commonly limit the model capacity in the current EEG-based emotion recognition models. Different from the existing transfer learning frameworks that only focus on feature separability on the source domain, we introduce an unsupervised adaptive pseudo labeling method to generate pseudo labels for the target domain and consider the feature separability in both the source domain and the target domain through an end-to-end domain adversarial training, where the model effectiveness and generalizability could be further enhanced. (2) We develop a prototypical learning-based adversarial discriminative domain adaptation method to explore latent variables of EEG signals corresponding to emotion categories, encode the semantic structure of EEG data, and learn subject-generalized prototypical representations for emotion representation across individuals. The characterized prototypical representations show a high feature concentration within one single emotion category and a high feature separability across different emotion categories. (3) We first formulate emotion recognition as a pairwise learning problem to replace the traditional classifier and greatly alleviate the label dependence on high-precision emotion labels. The pairwise learning provides us with an alternative way to interpret the EEG signals belonging to the same emotion category under a noisy labeling environment.

II. RELATED WORK

The existing EEG-based emotion recognition models with the transfer learning algorithms can be generally divided into two main categories.

(a) *Non-Deep Transfer Learning Models*: Pan et al. [20] proposed a transfer component analysis (TCA) algorithm to reduce the marginal distribution difference between the source and target domains, in which the transfer information was learned in a reproducing kernel Hilbert space through maximizing mean discrepancy. Zheng and Lu [21] introduced two types of subject-to-subject transfer methods to deal with the challenge of individual differences in EEG signal processing. One was to explore a shared common feature space underlying source and target domains using TCA and kernel principal analysis (KPCA), and another was to construct multiple personalized classifiers on the source domain and map the classifier parameters to the target domain using transductive parameter transfer (TPT). These non-deep transfer learning strategies have shown the possibility to bridge the discrepancy across two domains

with improved performance on the target domain. However, due to the small capacity and low complexity of modeling, the model accuracy and stability are still constrained, failing to meet the needs of affective brain-computer interfaces (aBCIs) in real-world applications.

(b) Deep Transfer Learning Models: Most of the existing EEG-based emotion recognition models are based on deep transfer learning methods built with domain-adversarial neural network (DANN) proposed in [22]. The main idea of DANN is to find a shared feature representation for source and target domains with indistinguishable distribution differences and also maintain the predictive ability of the estimated features on the source samples for a specific classification task. Jin et al. [23] was the first to introduce DANN in aBCIs. Benefiting from the powerful feature representation ability of deep networks and the high efficiency of adversarial learning in distribution adaptation, the results have demonstrated that DANN based aBCI system is superior to the other existing methods. The current aBCI systems could be considered as a collection of DANN-based models, which typically start from three directions to enhance the DANN performance in resolving the EEG-based emotion recognition tasks.

- *Incorporating the prior knowledge of neuroscience and brain anatomy with DANN:* Inspired by the neuroscience findings of the asymmetry property between the left and right hemispheres in emotional responses, Li et al. [24] proposed a bi-hemisphere domain adversarial neural network (BiDANN), in which a global and two local domain discriminators were designed to learn discriminant features from each cerebral hemisphere related to emotion perception and to improve the feature stability to the variation of different domains. The experiments on the SEED database demonstrated that BiDANN achieved higher emotion recognition performance than the original DANN. Considering the emotional responses from different brain regions would be varied, Li et al. [6] introduced a region-global spatial-temporal neural network (R2G-STNN) to integrate the spatial-temporal information from local and global brain regions under importance guidance and characterize hierarchical feature representations for modeling. Similarly, under an assumption that not all EEG channels are equally important in emotion recognition tasks, Du et al. [25] integrated attention mechanism, long-short-term memory (LSTM), and DANN to propose an attention-based LSTM with domain discriminator (ATDD-LSTM), in which the nonlinear relations among different EEG channels were characterized in a data-driven approach and the informative emotion-related channels were optimally selected.
- *Incorporating the probability distribution with DANN:* To deal with the training instability of DANN, Luo et al. [26] introduced Wasserstein generative adversarial network with gradient penalty (WGAN-GP) to narrow down the distance between the marginal probability distributions of different subjects. The results showed the model stability was improved and a better EEG-based cross-subject emotion recognition was achieved. However, WGAN-GP

only considered the marginal distribution differences but ignored the joint distribution differences of different domains. To address this problem, Li et al. [27] introduced a joint domain adaptation network (JLNN), where the joint distribution adaptation (JDA) method was incorporated with a unified framework of task-invariant features (MDA) and task-specific features (CDA).

- *Incorporating the metric learning with DANN:* Wang et al. [28] incorporated an efficient prototype-based data representation to learn a more discriminative embedded feature space for EEG-based emotion recognition, in which the euclidean metric was adopted to assess the associations between the data samples and all the selected prototypes. In order to jointly estimate the feature separability and distribution alignment, Wang et al. [29] presented a prototype-based symmetric positive definite (SPD) matrix network to integrate DANN with the metric learning based on the Riemannian distance. Shen et al. [30] proposed a coupled projection transfer metric learning (CPTML) model to take the domain alignment and graph-based metric learning into account at the same time.

Although the above models have achieved higher accuracies compared to the original model with DANN in EEG-based emotion recognition tasks, there still exist three major technical challenges. First, the learned feature representation is susceptible to noise interference from both source and target domains and would further affect the model generalizability [31], [32]. Second, the current DANN-based models primarily concern the emotion classification loss on the source domain, which might easily lead to the over-fitting of source domain data and a decline in classification ability on target domain data. Third, the existing algorithms largely rely on a large amount of accurately labeled data. However, in practical EEG applications, it is difficult to collect high-precision labels for each single EEG trial.

III. METHODOLOGY

Suppose the EEG trials on the source domain $\mathcal{S}$ and target domain $\mathcal{T}$ are given as $D_s = \{X_s, Y_s\}$ and $D_t = \{X_t, Y_t\}$, where $\{X_s, Y_s\} = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ and $\{X_t, Y_t\} = \{(x_i^t, y_i^t)\}_{i=1}^{N_t}$. Here, X_i^s and X_i^t are EEG samples, and Y_i^s and Y_i^t are the corresponding emotion labels. To make the narrative clearer, the frequently used notations are summarized in Table I. As shown in Fig. 1, the proposed PR-PL includes two main parts (prototypical representation and pairwise learning), with three loss functions (domain adversarial loss, supervised pairwise loss (source domain), and unsupervised pairwise loss (target domain)). In the prototypical representation, three types of features are defined. The characterized features $f(X_s)$ or $f(X_t)$ from the EEG samples are termed as *sample features*, which are forced to be as indistinguishable as possible from the source and target domains. Under an assumption that any emotion could be represented by a prototype via prototypical learning, the *prototype features* of each emotion category are learned based on the sample features $f(X_s)$ and Y_s from the source domain. The interaction relationships between the sample features and prototype features are measured and the *interaction features* are characterized which

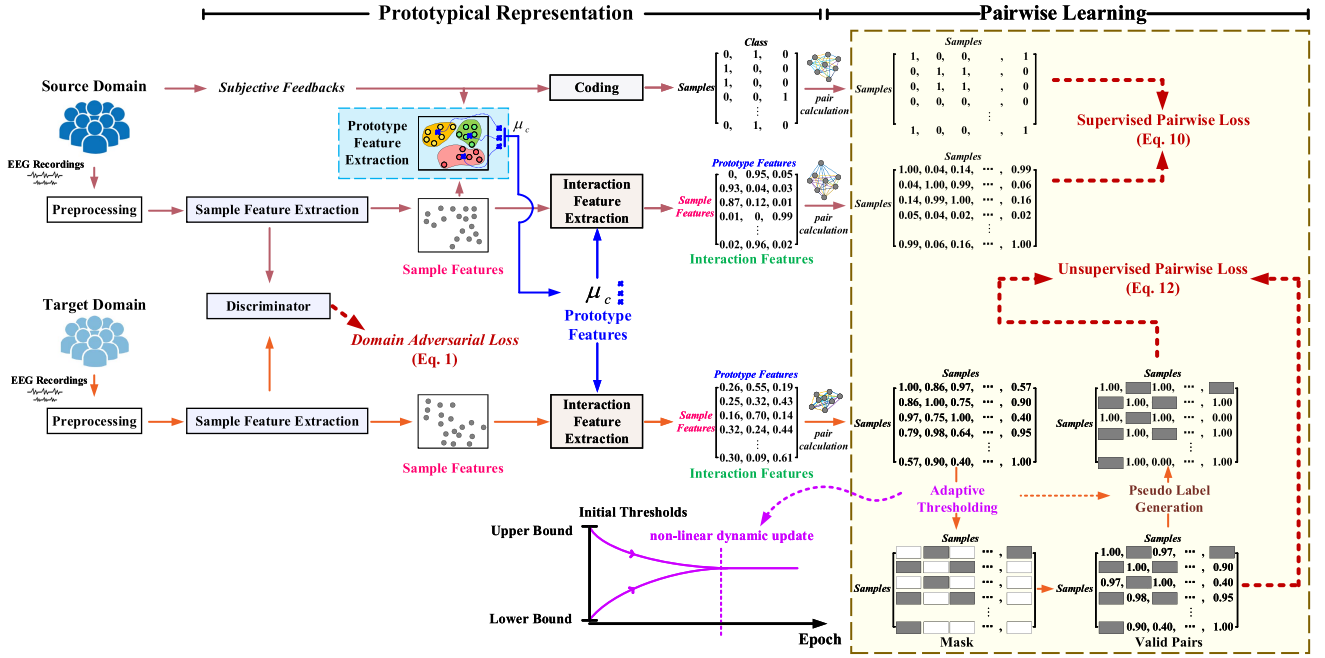


Fig. 1. The proposed PR-PL framework. In this proposed PR-PL, the domain adversarial loss, supervised pairwise loss, and unsupervised pairwise loss are defined in (1), (10), and (12), respectively. The final objective function is a combination of the above three loss functions, given in (17).

TABLE I
FREQUENTLY USED NOTATIONS AND DESCRIPTIONS

Notation	Description
$\mathcal{S} \setminus \mathcal{T}$	source \ target domain
$x^s \setminus x^t$	source \ target feature
$y^s \setminus y^t$	source \ target label
$N^s \setminus N^t$	number of source \ target samples
$D^s \setminus D^t$	the source \ target dataset
$f(\cdot)$	sample feature extractor
$d(\cdot)$	domain discriminator
$h(\cdot)$	bilinear operation
μ_c	prototype features
l_i	interaction features
θ_f	the parameter of the feature extractor $f(\cdot)$
θ_d	the parameter of the discriminator $d(\cdot)$
S	bilinear transformation matrix in $h(\cdot)$

will be used in the following pairwise learning. In pairwise learning, the pair relationships on both source and target domains are explored. As the information about Y_t on the target domain is unseen during model training, an unsupervised pairwise learning strategy is proposed for pseudo labeling and mask generation with a non-linear dynamic thresholding method.

A. Sample Feature Extraction

To make both source and target domain data satisfy the assumption of independent and identical distribution and obey the same distribution, we characterize the sample features based on a domain adversarial training introduced in DANN. Here, the distribution difference between the source domain and the target domain is alleviated and the sample features with domain invariant properties are characterized. Through this process, the individual differences in EEG signals could be alleviated and

the generalization ability of the models could be improved [6], [24], [25], [26]. Specifically, the domain difference between the sample feature $f(X_s)$ on the source domain and the sample feature $f(X_t)$ on the target domain are minimized by adopting domain adversarial training. Here, $f(\cdot)$ with the parameter θ_f is the designed feature extractor for extracting the sample features from EEG signals. A domain discriminator $d(\cdot)$ with the parameter θ_d is introduced to distinguish whether the characterized sample features ($f(X_s)$ or $f(X_t)$) come from the source domain ($\mathcal{S}$) or the target domain ($\mathcal{T}$), whose loss function is a standard two-category cross-entropy loss function, given as

$$\mathcal{L}_{disc}(\theta_f, \theta_d) = - \sum_{i=1}^{N_s} \log d(f(x_i^s)) - \sum_{i=1}^{N_t} \log (1 - d(f(x_i^t))), \quad (1)$$

which is termed as *domain adversarial loss* in Fig. 1.

In the training process, we adopt the end-to-end training method [22], and implement the domain adversarial training by introducing a gradient reversal layer. The feature extractor $f(\cdot)$ maximizes the classification ability to enhance emotion recognition performance and at the same time, the discriminator $d(\cdot)$ minimizes the domain discrimination to reduce the distribution difference between the source and target domains. The objective function in the domain adversarial training process could be represented as

$$\min_{\theta_f} \max_{\theta_d} \mathcal{L}_{classifier}(\theta_f) - \lambda \mathcal{L}_{disc}(\theta_f, \theta_d). \quad (2)$$

Here, $\mathcal{L}_{\text{classifier}}(\theta_f)$ is the classification loss to measure the modeling capability on the emotion recognition task. In this paper, $\mathcal{L}_{\text{classifier}}(\theta_f)$ will be realized by supervised pairwise learning on the source domain and unsupervised pairwise learning on the target domain introduced in Sections III-D and III-E below. $\mathcal{L}_{\text{disc}}(\theta_f, \theta_d)$ is the adversarial loss for the discriminator given in (1) to be trained to distinguish the sample features characterized from source and target domains. θ_f and θ_d are the parameters of $f(\cdot)$ and $d(\cdot)$. λ is a balanced hyperparameter for ensuring the stability of domain adversarial, which is given by an exponential growth method as

$$\lambda = \frac{2}{1 - \exp(-p)} - 1. \quad (3)$$

Here, p is a factor related to the training round, given by a ratio of the current training round to the maximum training round.

B. Prototype Feature Extraction

We assume that there exists a prototype representation for each emotion category. Through prototypical learning, the prototype features could be learned to indicate the representation property of every single emotion category. Based on the sample features extracted from different subjects under different emotions, we could consider these sample features are distributed around the prototype features. In other words, for each emotion category, the prototype feature could be considered the “center of mass” of all the sample features. From the perspective of a probability distribution, the prototype feature of an emotion category can be regarded as the mean value of the sample features of that emotion, and the variance of the distribution could be caused by the unstable EEG patterns, including but not limited to individual differences and the uncertainty of emotion expression. Assume that the sample features under an emotion category c obey the Gaussian distribution $N(\mu_c, \sigma_c^2)$. The prototype feature of the emotion category c could be calculated as the mean vector μ_c of the sample feature distribution. For the source domain $\{X_s, Y_s\} = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$, the corresponding sample features are characterized by the feature extractor $f(\cdot)$ defined in Section III-A, given as $f(x_i^s)$. The prototype feature vector of the emotion category c is estimated as

$$\mu_c = \frac{1}{|X_s^c|} \sum_{x_i^s \in X_s^c} f(x_i^s), \quad (4)$$

where $X_s^c = \{(x_i^s, y_i^s = c)\}_{i=1}^{|X_s^c|}$ is a collection of source domain data belonging to the emotion category c in the current mini-batch, and $|X_s^c|$ is the corresponding sample size. In order to achieve optimal prototype representation, μ_c would undergo iterative updates during the model training process. Here, μ_c could be expressed as the centroid of the sample features of X_s^c . The mean value calculation is a widely used, simple, and effective noise reduction strategy, which can make the prototype features stronger than the sample features and help to alleviate the problem of the traditional DANN network being susceptible to related noise interference [33]. It is noted that we only use the source domain here for prototype feature extraction, where the emotional label information is adopted. After the model training,

the optimal prototype features of each emotion category are obtained, which would then be fixed during the model validation procedure.

C. Interaction Feature Extraction

In the traditional DANN feature extraction, the derived domain-invariant shared feature representation could be easily contaminated by the shared noises on the source and target domains. In this paper, we introduce a bilinear transformation to quantify the interactions between the sample features and the prototype features and then extract the interaction features for subsequent pairwise learning.

For a given d -dimensional sample feature $f(x_i)$ (x_i could either belong to the source domain or the target domain), its interactions with the prototype feature μ_c of the c th emotion category is measured by a bilinear transformation $h(\cdot)$, defined as

$$h(f(x_i), \mu_c) = f(x_i)^T S \mu_c, \quad (5)$$

where $S \in \mathbb{R}^{d \times d}$ is a random-initialized bilinear transformation matrix with trainable parameters and is not restricted by symmetry or positive definiteness. The bilinear transformation matrix S continues to be updated during the model learning process through a gradient backpropagation by optimizing the final objective function given in (17). If we remove the bilinear transformation matrix S from (5) and replace it with an identity matrix, then (5) could be considered as an inner product between the sample feature and the prototype feature. Compared to the inner product, a trainable transformation matrix could be more beneficial to enhance the feature representation capability for the downstream tasks, which is also evidenced in the ablation study.

Further, for a total of n emotion categories, the interaction feature of x_i is a feature vector consisting of the interactions between the corresponding sample feature $f(x_i)$ and each of the prototype features μ_c ($c = 1, \dots, n$), specified as

$$l_i = \text{softmax}([h(f(x_i), \mu_1), \dots, h(f(x_i), \mu_n)]), \quad (6)$$

where $\{\mu_1, \dots, \mu_n\}$ are the prototype features corresponding to the n emotion categories, based on the definition given in (4). To further enrich the non-linear characteristics of the interaction features and enhance the feature separability for achieving advanced emotion recognition performance, the softmax function is adopted here to strengthen the feature representation with prediction ability. Overall, l_i could be interpreted as the characterized interaction feature for mapping to a non-linear informative feature space by measuring the interaction relationship between the sample features and all the available prototype features.

D. Supervised Pairwise Learning on Source Domain

For the source domain, the traditional pointwise learning algorithms often regard the feature vector l_i^s as the predicted label of the sample feature x_i^s and use the cross-entropy loss function to match the label y_i^s of x_i^s based on supervised training. Then, if the sample features of one EEG trial match the prototype features of the c th emotion category the most, the EEG trial

will be assigned to the c th emotion category. However, this type of pointwise learning only focuses on the relationship between sample features and prototype features and ignores the relationship among different pairs of samples. For example, the samples belonging to different emotion categories should be separated from each other, and the samples belonging to the same emotion category should be gathered together.

Considering this, we introduce pairwise learning to replace the traditional pointwise learning and conduct pair calculation to capture the inherent relationship of different pairs of samples in terms of the interaction features obtained in (6). The pairwise loss function on the source domain could be defined as

$$\mathcal{L}_{\text{class}}(\theta) = \frac{1}{N_s N_s} \sum_{i,j \in N_s} L(r_{ij}^s, g(x_i^s, x_j^s; \theta)). \quad (7)$$

In the training process, the label information of source domain data is used to define r_{ij}^s in a supervised manner. Based on the given information of y_i^s and y_j^s , if $y_i^s = y_j^s$, then $r_{ij}^s = 1$; otherwise $r_{ij}^s = 0$. A supervised r_{ij}^s can ensure the stability of the training process and the generalization ability of the model. On the other hand, $g(x_i^s, x_j^s; \theta)$ is a similarity measurement of the samples x_i^s and x_j^s , with the parameter of θ . The loss function $L(\cdot)$ is a difference calculation of r_{ij} and $g(x_i^s, x_j^s; \theta)$, given by a two-category cross-entropy loss as

$$\begin{aligned} L(r_{ij}^s, g(x_i^s, x_j^s; \theta)) = & -r_{ij}^s \log(g(x_i^s, x_j^s; \theta)) \\ & - (1 - r_{ij}^s) \log(1 - g(x_i^s, x_j^s; \theta)), \end{aligned} \quad (8)$$

where $g(x_i^s, x_j^s; \theta)$ is a similarity measurement between x_i^s and x_j^s in terms of the characterized interaction features l_i^s and l_j^s as

$$g(x_i^s, x_j^s; \theta) = \frac{l_i^s \cdot l_j^s}{\|l_i^s\|_2 \|l_j^s\|_2}, \quad (9)$$

where $\cdot$ refers to inner product operation. A norm restriction $l/\|l\|_2$ is adopted here to make the similarity results located in the range of $[0, 1]$ and extract better and more robust feature representations for subsequent classification [34].

Overall, the objective function of the supervised pairwise learning on the source domain is defined as

$$\mathcal{L}_{\text{pairwise}}^s(\theta) = \frac{1}{N_s N_s} \sum_{i,j \in N_s} L(r_{ij}^s, g(x_i^s, x_j^s; \theta)) + \beta \mathcal{R}, \quad (10)$$

which is termed as *supervised pairwise loss* in Fig. 1. Here, $\theta = \{\theta_f, S\}$, θ_f is the parameter of feature extractor $f(\cdot)$, and S is the defined bilinear transformation matrix for interaction feature extraction. Besides, to avoid redundant feature extraction, a soft regularization $\mathcal{R}$ is introduced with a weight parameter of β , which is defined as

$$\mathcal{R} = \|P^T P - I\|_F. \quad (11)$$

Here, each row of the matrix P refers to the prototype feature belonging to one emotion category, $\|\cdot\|_F$ is a F norm of the matrix, and I is an identity matrix.

The above loss function (10) could be interpreted as a clustering loss, instead of an emotion category classification loss. The main optimization goal is to gather the EEG samples that

may belong to the same emotion category and separate the EEG samples that do not belong to the same category.

E. Unsupervised Pairwise Learning on Target Domain

In this paper, we also introduce unsupervised pairwise learning to improve the feature separability on the target domain, as

$$\mathcal{L}_{\text{pairwise}}^t(\theta_f, S) = \frac{1}{N_t N_t} \sum_{i,j \in N_t} L(r_{ij}^t, \frac{l_i^t \cdot l_j^t}{\|l_i^t\|_2 \|l_j^t\|_2}), \quad (12)$$

which is termed as *unsupervised pairwise loss* in Fig. 1. Here, l_i^t and l_j^t are the interaction features of x_i^t and x_j^t on the target domain, characterized by (6). The scalar r_{ij}^t symbolizes the pairing relationship of x_i^t and x_j^t . Since the label information of the target domain is completely unknown in the training process, we cannot accurately obtain the pairing relationship as the source domain. To address this issue, we introduce an adaptive thresholding method to generate a mask for selecting valid pair relationships and producing the corresponding pseudo labels. Suppose that r_{ij}^t is defined as

$$r_{ij}^t = \begin{cases} 1, & \text{if } l_i^t \cdot l_j^t \geq \tau_h \\ 0, & \text{if } l_i^t \cdot l_j^t < \tau_l, \end{cases} \quad i, j = 1, \dots, N_t, \quad (13)$$

where τ_h and τ_l are the upper and lower bounds to select valid pairs with high confidence for unsupervised pairwise learning (mask generation). Here, if the calculated pairwise similarity is higher or equal to the defined upper bound (τ_h), then the corresponding pseudo label of r_{ij}^t would be assigned to 1; while if the calculated pairwise similarity is lower than the defined lower bound (τ_l), then the corresponding pseudo label of r_{ij}^t would be assigned to 0. For the other pairs that do not meet the threshold requirement, we would consider that the model is uncertain about whether the sample pair is paired or not and consider these pairs as invalid results. In order to prevent incorrect optimization, the invalid pairs would be temporarily excluded from training and will not participate in the loss calculation at the current training round.

Considering the classification performance on the target domain could be quite unstable in the early training stage, we set a strict upper threshold (τ_h) and a lower threshold (τ_l) and exclude most of the pair results on the target domain to ensure the training stability. Along with the enhancement of model performance on the target domain, we can gradually lower the upper threshold and raise the lower threshold and allow more samples to participate in the training part. In other words, along with the increase in training steps, more pair relationships on the target domain could be included for model learning. Here, we form a non-linear dynamic update for adaptive thresholding, as

$$\tau_h^t = \tau_h^{t-1} - \frac{\tau_h^{t-1} - \tau_l^{t-1}}{\text{maxepoch}}, \quad (14)$$

$$\tau_l^t = \tau_l^{t-1} + \frac{\tau_h^{t-1} - \tau_l^{t-1}}{\text{maxepoch}}, \quad (15)$$

where τ_h^t and τ_l^t represent the upper and lower thresholds at the current training round t , respectively. *maxepoch* is the

maximum training round. Based on the given initial values τ_h^0 and τ_l^0 , the calculations of τ_h^t and τ_l^t could be considered as non-linear updates with respect to the training rounds as

$$\begin{cases} \tau_h^t = \tau_h^0 - \frac{\tau_h^0 - \tau_l^0}{2} \times \left(1 - \left(1 - \frac{2}{maxepoch}\right)^t\right) \\ \tau_l^t = \tau_l^0 + \frac{\tau_h^0 - \tau_l^0}{2} \times \left(1 - \left(1 - \frac{2}{maxepoch}\right)^t\right). \end{cases} \quad (16)$$

Overall, the final objective function of the proposed PR-PL (2) could be written as a combination of the domain adversarial loss (1), the supervised pairwise loss on the source domain (10), and the unsupervised pairwise loss on the target domain (12), given as

$$\min_{\theta_f, S} \max_{\theta_d} \mathcal{L}_{\text{pairwise}}^s(\theta_f, S) + \gamma \mathcal{L}_{\text{pairwise}}^t(\theta_f, S) - \lambda \mathcal{L}_{\text{disc}}(\theta_f, \theta_d), \quad (17)$$

where γ is a hyperparameter to control the importance of the unsupervised pairwise loss on the target domain, defined as

$$\gamma = \delta \times \frac{epoch}{maxepoch}, \quad (18)$$

where $epoch$ is the current training round, and $maxepoch$ is the maximum training round. δ is a constant value. During the model training, we compute the final objective function on the basis of the data samples $(\{x^s, y^s\}, \{x^t, -\})$ from one random mini-batch with the stochastic gradient descent for parameter updating.

IV. EXPERIMENTAL RESULTS

A. Emotional Databases and Data Preprocessing

We validate the proposed PR-PL on two well-known public databases: SEED [13] and SEED-IV [35]. In SEED database [13], a total of 15 subjects were invited. Each subject performed three sessions on different days and each session contained 15 trials. A total of three emotions were elicited (negative, neutral, and positive). In SEED-IV database [35], a total of 15 subjects participated in the experiment. For each subject, a total of 3 sessions were performed on different days and each session contained 24 trials. A total of four emotions were elicited (happiness, sadness, fear, and neutral). The EEG signals of both SEED and SEED-IV databases were simultaneously collected using the 62-channel ESI Neuroscan system.

For EEG preprocessing, the data sampling rate was first downsampled to 200 Hz, and the contaminated noises (e.g., EMG and EOG) were manually removed. Then, the data were filtered by a band-pass filter of 0.3 Hz to 50 Hz. For each trial, the data was divided into a number of segments with a length of 1 s. Based on the pre-defined five frequency bands: Delta (1-3 Hz), Theta (4-7 Hz), Alpha (8-13 Hz), Beta (14-30 Hz), and Gamma (31-50 Hz), the corresponding differential entropy (DE) features were extracted to represent the logarithm energy spectrum in a specific frequency band and total 310 features (5 frequency bands $\times$ 62 channels) were obtained for one EEG segment. Then, all the features in a trial were smoothed with the linear dynamic system (LDS) method, which can utilize the time dependency of emotion changes and filter out emotion unrelated

and noisy EEG components [36]. During the cross-validation, there would not exist any information leakage problem, since the source domain data and the target domain data are obtained from distinct trials. Here, the EEG data preprocessing adheres to the same standards as the prior studies for a fair comparison with the literature.

B. Implementation Results

In our experiments, the feature extractor $f(\cdot)$ and the discriminator $d(\cdot)$ are both made up of the multilayer perceptron (MLP) with the Relu activation function. All the parameters are randomly initialized from a uniform distribution. The bilinear operator matrix S given in (5) is also randomly initialized, with the size of 64×64 . In the model architecture, the feature extractor structure is designed as 310 (input layer)-64 (hidden layer 1)-Relu activation-64 (hidden layer 2)-Relu activation-64 (output feature layer). The discriminator structure is designed as 64 (input layer)-64 (hidden layer 1)-Relu activation-dropout layer-64 (hidden layer 2)-1 (output layer)-Sigmoid activation. Besides, we adopt an RMSprop optimizer for network training, which shows a better performance than the other classic optimizers. The learning rate is set to 1e-3 and the mini-batch size for training is 96. To avoid overfitting problems, we use L2 regularization with the weight of 1e-5 in the network. The β value in (10) is assigned as 0.01, and the δ value in (18) is set to 2. The threshold τ_h^0 and τ_l^0 are initialized as 0.9 and 0.5 respectively. All the models are trained on an NVIDIA GeForce RTX 2080 GPU, with CUDA 10.0 using the Pytorch API. It is notably that, in the model training process, only the raw target domain data are used without any label information involved, which is the same as the previous EEG-based emotion recognition methods using transfer learning framework [5], [8], [37], [38].

C. Experiment Protocols

To make a thorough comparison with the literature, we validate the proposed PR-PL using four different cross-validation evaluation protocols under consideration of the variants in subjects and sessions. (1) *Within-subject cross-session cross-validation*. For one subject's data, the first two sessions are used as the source, and the latter session is used as the target. The average accuracies and standard deviations across subjects are calculated as the final results. (2) *Within-subject single-session cross-validation*. Following the cross-validation evaluation protocol presented in the existing studies [13], [35], for each session of one subject, we use the first 9 (SEED) or 16 (SEED-IV) trials as the source and the rest 6 (SEED) or 8 (SEED-IV) trials as the target. The results are reported as the average performance across all the subjects. The above two cross-validation evaluation protocols are subject-dependent to estimate the model stability within one subject. To further evaluate the robustness and stability of the proposed PR-PL across different subjects, we also perform the model validation on the following two evaluation protocols (subject-independent). (3) *Cross-subject cross-session leave-one-subject-out cross-validation*. To estimate the model efficacy on the unknown subject(s) and session(s), we use one

TABLE II
MODEL PERFORMANCE ON SEED USING WITHIN-SUBJECT CROSS-SESSION
CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
<i>Traditional machine learning methods</i>			
RF* [40]	76.42±11.15	KNN* [41]	75.68±13.82
TCA* [20]	74.27±12.88	CORAL* [42]	84.18±09.81
SA* [43]	69.84±09.46	GFK* [44]	78.79±09.39
<i>Deep learning methods</i>			
DAN* [45]	89.16±07.90	SimNet* [33]	86.88±07.83
DDC* [46]	91.14±05.61	ADA [27]	89.13±07.13
DANN* [22]	89.45±06.74	MMD [27]	84.38±12.05
JDA-Net [27]	91.17±08.11	DCORAL* [47]	88.67±06.25
PR-PL	93.18±06.55		

subject's all sessions' data as the target and the remaining subjects' all sessions as the source. Take the SEED database as an example. Based on a total of 3394 samples in a single session for each subject, the data size of the source domain in the cross-subject cross-session is 14 subjects $\times$ 3 sessions $\times$ 3394 samples, and that of the target domain is 1 subject $\times$ 3 sessions $\times$ 3394 samples. We repeat the training validation until each subject is treated as the target once and report the average performance across all the subjects as the final results. (4) *Cross-subject single-session leave-one-subject-out cross-validation*. It is the most widely used validation scheme in EEG-based emotion recognition tasks [5], [26], [27], [39]. One subject's single session's data is treated as the target and the other remaining subjects' single session's data is used as the source. Take the SEED database as an example. Based on a total of 3,394 samples in a single session for each subject, the data size of the source domain in the cross-subject single-session is 14 subjects $\times$ 1 session $\times$ 3,394 samples, and that of the target domain is 1 subject $\times$ 1 session $\times$ 3,394 samples. We repeat the training validation until each subject is treated as the target once and report the average performance across all the subjects as the final results. Here, same as the other existing studies, we only consider the first session in the cross-subject single-session cross-validation.

In the following, the model performance under the aforementioned four different cross-validation evaluation protocols is reported and compared with the existing literature. Here, the model results reproduced by us are indicated by *.

D. Within-Subject Cross-Session Cross-Validation Results

We evaluate the model performance on each subject's data across all sessions. The final experimental results are the average accuracies and standard deviations across subjects, as shown in Table II (SEED) and Table III (SEED-IV). Compared to the experimental results presented in the existing traditional machine learning methods and deep learning methods, the proposed PR-PL achieves superior performance in both SEED and SEED-IV databases, where the results are 93.18%±6.55% for SEED (three-class emotion recognition) and 74.62%±14.15% for SEED-IV (four-class emotion recognition).

TABLE III
MODEL PERFORMANCE ON SEED-IV USING WITHIN-SUBJECT CROSS-SESSION
CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
<i>Traditional machine learning methods</i>			
RF* [40]	60.27±16.36	KNN* [41]	54.18±16.28
TCA* [20]	51.88±15.84	CORAL* [42]	66.06±15.13
SA* [43]	52.81±09.53	GFK* [44]	56.14±12.15
<i>Deep learning methods</i>			
DCORAL [48]	65.10±13.20	DAN [48]	60.20±10.20
DDC [48]	68.80±16.60	MEERNet [48]	72.10±14.10
PR-PL	74.62±14.15		

TABLE IV
MODEL PERFORMANCE ON SEED USING WITHIN-SUBJECT SINGLE-SESSION
CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
<i>Traditional machine learning methods</i>			
SVM [50]	83.99±09.72	GRSLR [52]	87.39±08.64
RF [50]	78.46±11.77	GSCCA [53]	82.96±09.95
CCA [50]	77.63±13.21	DBN [13]	86.08±08.34
<i>Deep learning methods</i>			
DGCNN [51]	90.40±08.49	RGNN [8]	94.24±05.95
ATDD-DANN [25]	91.08±06.43	BiHDM [50]	93.12±06.06
R2G-STNN [6]	93.38±05.96	SimNet* [33]	90.13±10.84
BiDANN [24]	92.38±07.04	STRNN [54]	89.50±07.63
GCNN [50]	87.40±09.20	DANN [50]	91.36±08.30
PR-PL	94.84±09.16		

TABLE V
MODEL PERFORMANCE ON SEED-IV USING WITHIN-SUBJECT
SINGLE-SESSION CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
<i>Traditional machine learning methods</i>			
SVM [50]	56.61±20.05	GRSLR [52]	69.32±19.57
RF [50]	50.97±16.22	GSCCA [53]	69.08±16.66
CCA [50]	54.47±18.48	DBN [13]	66.77±07.38
<i>Deep learning methods</i>			
DGCNN [51]	69.88±16.29	RGNN [8]	79.37±10.54
GCNN [50]	68.34±15.42	BiHDM [50]	74.35±14.09
A-LSTM [50]	69.50±15.45	SimNet* [33]	71.38±13.12
BiDANN [24]	70.29±12.63	DANN [50]	63.07±12.66
PR-PL	83.33±10.61		

E. Within-Subject Single-Session Cross-Validation Results

To evaluate the model effectiveness on the data that are collected from the same subject on the same day (session), we conduct within-subject single-session cross-validation as the previous studies [6], [8], [24], [49], [50]. Tables IV and V report the corresponding experimental results on SEED and SEED-IV databases, respectively. Compared to the existing methods (such as BiDANN [24], DGCNN [49], BiHDM [50], RGNN [8]), the proposed PR-PL achieves promising results on both two benchmark databases (SEED: 94.84±09.16; SEED-IV: 83.33±10.61).

TABLE VI
MODEL PERFORMANCE ON SEED USING CROSS-SUBJECT CROSS-SESSION
LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
<i>Traditional machine learning methods</i>			
RF* [40]	69.60±07.64	KNN* [41]	60.66±07.93
SVM* [55]	68.15±07.38	Adaboost* [56]	71.87±05.70
TCA* [20]	64.02±07.96	CORAL* [42]	68.15±07.83
SA* [43]	61.41±09.75	GFK* [44]	66.02±07.59
<i>Deep learning methods</i>			
DCORAL* [47]	81.97±05.16	DAN* [45]	81.04±05.32
DDC* [46]	82.17±04.96	DANN* [22]	81.08±05.88
PR-PL	85.56±04.78		

TABLE VII
MODEL PERFORMANCE ON SEED-IV USING CROSS-SUBJECT CROSS-SESSION
LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
<i>Traditional machine learning methods</i>			
RF* [40]	50.98±09.20	KNN* [41]	40.83±07.28
SVM [57]	51.78±12.85	Adaboost* [56]	53.44±09.12
TCA [50]	56.56±13.77	CORAL* [42]	49.44±09.09
SA [50]	64.44±09.46	GFK* [44]	45.89±08.27
KPCA [57]	51.76±12.89	DNN [57]	49.35±09.74
<i>Deep learning methods</i>			
DGCNN [49]	52.82±09.23	DAN [57]	58.87±08.13
RGNN [8]	73.84±08.02	BiHDM [50]	69.03±08.66
BiDANN [24]	65.59±10.39	DANN [57]	54.63±08.03
PR-PL	74.92±07.92		

F. Cross-Subject Cross-Session Leave-One-Subject-Out Cross-Validation Results

Considering the possible real-world application scenarios in aBCI, we further validate the model performance when the data are from different subjects and collected on different days (sessions). As reported in Tables VI and VII, the emotion recognition performance of the proposed PR-PL is $85.56\% \pm 4.78\%$ for three-class classification task on SEED and $74.92\% \pm 7.92\%$ for four-class classification task on SEED-IV. Due to the variants in subjects and sessions, this evaluation protocol poses the greatest challenge to the model's effectiveness in the EEG-based emotion recognition tasks among the four different cross-validation evaluation protocols (within-subject cross-session: variant in session only; within-subject single-session: n/a; cross-subject cross-session: variants in both subjects and sessions; cross-subject single-session: variants in subjects only). Compared to the existing studies, the proposed PR-PL increases the classification accuracy to 3.39% and 1.08% for SEED and SEED-IV, with smaller standard deviations. These results demonstrate the proposed PR-PL has good effectiveness and generalizability in addressing the variations among subjects and sessions for the EEG-based emotion recognition tasks.

TABLE VIII
MODEL PERFORMANCE ON SEED USING CROSS-SUBJECT SINGLE-SESSION
LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
<i>Traditional machine learning methods</i>			
TKL [24]	63.54±15.47	T-SVM [24]	72.53±14.00
TCA [50]	63.64±14.88	TPT [57]	75.17±12.83
KPCA [57]	61.28±14.62	GFK [50]	71.31±14.09
SA [50]	69.00±10.89	DICA [58]	69.40±07.80
DNN [57]	61.01±12.38	SVM [57]	58.18±13.85
<i>Deep learning methods</i>			
DGCNN [49]	79.95±09.02	DAN [57]	83.81±08.56
RGNN [8]	85.30±06.72	BiHDM [50]	85.40±07.53
WGAN-GP [26]	87.10±07.10	MMD [27]	80.88±10.10
ATDD-DANN [25]	90.92±01.05	JDA-Net [27]	88.28±11.44
R2G-STNN [6]	84.16±07.63	SimNet* [33]	81.58±05.11
BiDANN [24]	83.28±09.60	DResNet [58]	85.30±08.00
ADA [27]	84.47±10.65	DANN [27]	81.65±09.92
PR-PL	93.06±05.12		

TABLE IX
MODEL PERFORMANCE ON SEED-IV USING CROSS-SUBJECT SINGLE-SESSION
LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
<i>Traditional machine learning methods</i>			
GFHF [59]	49.29±07.60	GFK [59]	44.04±09.31
TCA [59]	37.01±10.47	JDA [59]	54.57±05.47
GAKT [59]	58.49±06.23	MIDA [59]	60.22±08.69
JAGP [59]	78.54±07.33	MS-STM [5]	61.41±09.72
<i>Deep learning methods</i>			
DGCNN [49]	68.73±08.34	JDA-Net [27]	70.83±10.25
UDDA [60]	73.14±09.43	MS-MDA [61]	59.34±05.48
DANN [60]	54.63±08.03	DAN [60]	58.87±08.13
WGAN-GP [26]	60.60±15.76	MWACN [62]	74.60±10.77
PR-PL	81.32±08.53		

G. Cross-Subject Single-Session Leave-One-Subject-Out Cross-Validation Results

The cross-subject single-session leave-one-subject-out cross-validation is the most popular evaluation protocol in current aBCI systems, which provides an objective measurement to evaluate the model stability when different subjects' data are considered. In Table VIII, we summarize the experimental results on the SEED database and compare them with the literature. The results show the proposed PR-PL achieves the best performance (93.06 ± 05.12), which leads 2.14% against the reported results in the literature. Compared to the latest proposed DANN-based deep transfer learning networks in which both source and target domain data are used for modeling (e.g., ATDD-DANN [25], R2G-STNN [6], BiHDM [50], BiDANN [24], and WGAN-GP [26]), the proposed PR-PL with pairwise learning can avoid the inherent defects of DANN design (e.g., only considers feature separability on source domain). Similar observations are also found in the SEED-IV results (Table IX), where the proposed PR-PL achieves superior performance with 6.72% enhancement.

TABLE X
THE ABLATION STUDY OF OUR PROPOSED MODEL

<i>Ablation study about training strategy</i>	<i>P_{acc}</i>
w/o discriminator and target information	83.30±04.21
w/o pairwise learning on the source and target	87.50±06.64
w/o pairwise learning on the target	88.81±06.63
w/o the bilinear transformation matrix <i>S</i>	88.31±04.42
w/o prototypical representation	91.00±04.65
w/o pseudo labeling method	92.13±05.90
<i>About hyperparameter controlling strategy</i>	<i>P_{acc}</i>
w/ fixed pseudo labeling	89.92±07.21
w/ linear dynamic pseudo labeling	92.46±04.95
w/ fixed γ for unsupervised pairwise loss in Eq. 17	89.47±10.22
PR-PL	93.06±05.12

V. DISCUSSION

We further evaluate the effect on model performance when different parameter settings are adopted. All the present results in this session are based on the SEED database using the commonly used cross-subject single-session leave-one-subject-out cross-validation evaluation protocol in the EEG-based emotion recognition tasks.

A. Ablation Study

We conduct the ablation study to systematically explore the effectiveness of different components in the proposed model and examine the corresponding contributions to the overall performance. As shown in Table X, it is found that the introduction of domain adversarial training can greatly enhance the emotion recognition performance on the target domain. When the model is without discriminator and target domain information, the recognition accuracy reduces from 93.06% to 83.30%. Such a significant drop shows the significant impact of individual differences problem on model performance and highlights the great potential of transfer learning in aBCI applications. Besides, the results show a combination of pairwise learning on the source and target domain benefits to the model performance, where the recognition accuracy is increased by 6.35% (from 87.50% to 93.06%). Without a consideration of the feature separability on the target domain (w/o pairwise learning on the target), the model performance would drop from 93.06% to 88.81%. In addition, the introduction of the bilinear transformation matrix *S* in (5) can improve the model performance, where the recognition accuracy could be increased by 4.75% (from 88.31% to 93.06%). In a comparison between with and without pseudo labeling methods, incrementally including the valid pairs from the target domain in the training process is helpful to the learning stability, as shown by the performance rise from 92.13% to 93.06%. Further, the modeling effectiveness with different pseudo labeling methods is examined. The corresponding accuracy increases from 89.92% to 92.46% when the pseudo labeling method changes from fixed to linear dynamic. The accuracy further increases to 93.06%, when a nonlinear dynamic-based adaptive pseudo labeling method is adopted. The results show a non-linear dynamic-based adaptive pseudo labeling could be helpful to screen out the valid paired samples and improve the model trainability. For the final loss function given in (17),

instead of using a fixed γ for the unsupervised pairwise loss on the target domain, we propose to update γ gradually along with the training epochs to prevent model learning failures in the early training stage and balance the relationships among different losses. The benefit of a dynamic γ is also reflected in the ablation study, where the recognition accuracy increases from 89.47% and 93.06%. All the reported results using different components in the proposed PR-PL exist significant differences, with a p-value smaller than 0.001.

B. Effect of Noisy Labels

To further verify the model robustness during noisy label learning, we randomly contaminate the source labels with $\eta\%$ noises and test the corresponding model performance on unknown target data. Specifically, we replace $\eta\%$ real labels in Y_s using randomly generated labels and train the model in a supervised learning manner. Then, we test the trained model performance on the target domain. Note here that the noisy contamination is only conducted on the source domain, as the target domain needs to be used for model evaluation. In the implementation, $\eta\%$ value is adjusted to 10%, 20%, and 30%, respectively. The corresponding model accuracies with the standard deviations are 89.22%±6.05%, 88.39%±6.73%, and 87.71%±5.02%. It shows that, with an increase of label noise ratio from 10% to 30%, the model performance decreases slightly, with a decrease rate of 1.69%. These results demonstrate the proposed PR-PL is a reliable model which has a better tolerance to noisy labels. In the future works, the recently proposed novel methods, such as [62] and [63], could be incorporated to further eliminate more general noises in EEG signals and improve the model stability in the cross-corpus applications.

C. Confusion Matrices

To qualitatively study the model performance in each emotion class, we visually analyze the confusion matrices and compare results with the latest models [8], [24], [50]. As shown in Fig. 2, it shows all the models are good at distinguishing positive emotions from other emotions (the recognition rates are all above 90%), but it is relatively poor at distinguishing negative and neutral emotions. For example, the recognition rate of neutral in BiDANN [24] is even lower than 80% (76.72%). Compared to the existing methods ((a), (b), and (c)), our proposed model can enhance the model recognition ability, especially for distinguishing neutral and negative emotions. As shown in (d), the recognition rates for negative, neutral, and positive emotions are 92.10%, 90.39%, and 96.50%. More confusion matrix results corresponding to the other cross-validation evaluation protocols such as within-subject cross-session, within-subject single-session, cross-subject cross-session leave-one-subject-out are reported in Appendix I of Supplementary Materials, available online.

D. Visualization of Learned Representation

To verify the effectiveness of the proposed model from a more intuitive perspective, we visualize the characterized sample and interaction features of source and target domains using

Negative	0.8459	0.1073	0.0467
Neutral	0.1839	0.7672	0.0489
Positive	0.0689	0.0241	0.9070
	Negative	Neutral	Positive
(a)			
Negative	0.8083	0.1178	0.0739
Neutral	0.0567	0.8450	0.0983
Positive	0.0724	0.0213	0.9063
	Negative	Neutral	Positive
(b)			
Negative	0.7914	0.1452	0.0634
Neutral	0.1441	0.8483	0.0076
Positive	0.0451	0.0382	0.9167
	Negative	Neutral	Positive
(c)			
Negative	0.9210	0.0668	0.0123
Neutral	0.0690	0.9039	0.0271
Positive	0.0194	0.0156	0.9650
	Negative	Neutral	Positive
(d)			

Fig. 2. Confusion matrices of different models. (a) BiDANN [24], (b) BiHDM [50], (c) RGNN [8], and (d) PR-PL.

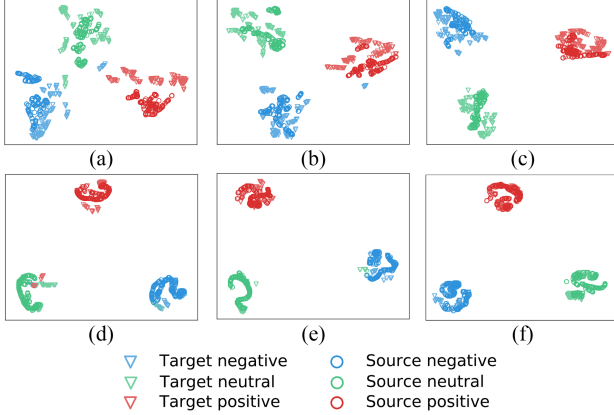


Fig. 3. T-SNE visualization of the learned features from the source and target domains under different model settings. (a), (b) and (c) are the sample features extracted by w/o pairwise learning on the source and target domains, w/o pairwise learning on the target domain, and PR-PL. (d), (e) and (f) are the interaction features extracted by w/o pairwise learning on the source and target domains, w/o pairwise learning on the target domain, and PR-PL.

T-SNE [64] in Fig. 3. Here, we randomly select 500 samples from the source domain and 500 samples from the target domain at one validation round for feature visualization. Compared to the representation learned by the other model settings (w/o pairwise learning on the source and target and w/o pairwise learning on the target), the representation learned by the proposed PR-PL forms more separated emotional clusters. Comparing the extracted sample features (c) and interaction features (f) by the proposed PR-PL, the separability of the extracted interaction features from different emotion classes is further enlarged and at the same time, the concentration of the feature distribution for each emotion is also improved. To better visualize the validity of the extracted features by the proposed PR-PL, a complete feature visualization in terms of the extracted features at the first hidden layer of the feature extractor $f(\cdot)$, sample features, prototype features, and interaction features at each validation round is reported in Appendix II of Supplementary Material, available online. Besides, to reveal the underlying emotion-related neural expression of the extracted features, we also conduct the topographic analysis on the obtained prototype features. Considering the paper length constraints, more implementation details with the corresponding results are presented in Appendix III of Supplementary Material, available online.

E. Computation Time and Incomplete Target Domain

To evaluate the time complexity of the proposed PR-PL, we estimate the computation time on all the target domain

 TABLE XI
MODEL PERFORMANCE WITH INCOMPLETE TARGET DOMAIN

Validation Set (K trials)	Test Set ($(N_{max} - K)$ trials)	P_{acc}
3	12	85.07 ± 05.84
6	9	85.35 ± 07.72
9	6	86.88 ± 07.90
12	3	88.58 ± 08.98

data. The average cost time of each data sample for emotion recognition is 0.00061 s, and the standard deviation is 0.00049 s. The transfer learning framework, on the other hand, aims to achieve knowledge transfer by adapting source knowledge to the unknown target domain. However, not all of the target domain data would be always accessible for model learning. Here, we further evaluate the robustness of the proposed PR-PL in the presence of an incomplete target domain. Specifically, we select the first K trials on the target domain as the validation set and the remaining trials ($N_{max} - K$) as the test set. For the SEED database, N_{max} is equal to 15. To measure the model stability under different incomplete conditions, we adjust the K value from 3 to 12. To avoid biased class distribution, the step of K value is set to 3 (three-class classification task). In the implementation, only the source domain and the target domain's validation set are used for model learning. Then, the learned model is applied to the target domain's test set. In other words, the test set used for model validation is completely unseen during model training. The corresponding results are reported in Table XI, which shows that the proposed PR-PL could still be effective even in the case of incomplete target domain data.

VI. CONCLUSION

The article proposes a novel transfer learning framework with prototypical representation-based pairwise learning (PR-PL), that characterizes EEG data with prototypical representations and formulates the EEG-based emotion recognition tasks as pairwise learning. To estimate the model sensitive to the variants in subjects and sessions, we evaluate the proposed PR-PL on two well-known emotional databases (SEED and SEED-IV) under four different cross-validation evaluation protocols (within-subject cross-session, within-subject single-session, cross-subject cross-session, and cross-subject single-session) and compare it with the existing methods. Our extensive experimental results show that the proposed PR-PL achieves promising results on all four cross-validation evaluation protocols. Further, the model stability and generalizability under

various experimental settings are fully validated. These experimental findings indicate the superiority of the proposed PR-PL in tackling individual differences and noisy labeling issues in aBCI systems.

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Rushuang Zhou is currently working toward the PhD degree with the Department of Biomedical Engineering, City University of Hong Kong, Hong Kong. His research interests include affective computing and transfer learning.



Zhiguo Zhang (Member, IEEE) received the BEng degree from Tianjin University, in 2000, the MPhil degree from the University of Science and Technology of China, in 2003, and the PhD degree from the University of Hong Kong, in 2008. He is now a professor with the Institute of Computing and Intelligence, Harbin Institute of Technology, Shenzhen, China. His research interests include biomedical signal processing, neural engineering, and brain-inspired computation.



Hong Fu received the PhD degree from The Hong Kong Polytechnic University, in 2007. She is now an assistant professor with the Department of Mathematics and Information Technology, The Education University of Hong Kong, Hong Kong. Her research interests include computer vision algorithms and artificial intelligence in healthcare and education.



Li Zhang received the PhD degree from the Department of Electrical and Electronics, University of Hong Kong, in 2017 and joined the School of Biomedical Engineering, Medical School, Shenzhen University, in 2018 as an associate researcher. His current research interests mainly focus on biomedical signal processing, numerical optimization, and imaging genetics.



Linling Li received the PhD degree from the University of Chinese Academy of Sciences, China, in 2017, and then received the postdoctoral training with Shenzhen University, China. She is now an associate researcher with the School of Biomedical Engineering, Medical School, Shenzhen University, China. Her current research interests focus on the development of EEG neurofeedback technique and related neuroimaging mechanisms.



Gan Huang received the PhD degree from Shanghai Jiao Tong University, China, in 2013, and then received his postdoctoral training with Université Catholique de Louvain in Belgium. He is now an assistant professor with the School of Biomedical Engineering, Shenzhen University, China. He has published several high-quality papers on journals including *Neuroimage*, *Neural Network*, and *Neurocomputing*. His current research interests focus on neural modulation, brain-computer interface and neuroprosthesis.



Fali Li received the PhD degree from the University of Electronic Science and Technology of China, in 2020. He is now an associate researcher with the School of Life Science and Technology, Center for Information in Medicine, University of Electronic Science and Technology of China, China. His research interests include brain-computer interface, bio-informatics, and neural engineering.



Yuan-Ting Zhang (Fellow, IEEE) is currently the chairman and director of Hong Kong Center for Cerebro-cardiovascular Health Engineering and the chair professor with the City University of Hong Kong. He is a LRG member of Karolinska Institutet MWLC. He was the sensing system architect in Health Technology and Sensing Hardware Divisions at Apple Inc., California, the founding director of the Key Lab for Health Informatics of Chinese Academy of Sciences (CAS) and the founding director of CAS-SIAT Institute of Biomedical and Health Engineering.

He dedicated his service to the Chinese University of Hong Kong from 1994 to 2015, where he served as the first head of the Division of Biomedical Engineering. He has been the editor-in-chief for IEEE Reviews in Biomedical Engineering since 2016 and chair of the IEEE 1708 Working Group. He was the editor-in-chief for *IEEE Transactions on Information Technology in Biomedicine* and the first editor-in-chief of *IEEE Journal of Biomedical and Health Informatics*. He served as vice preside of *IEEE Engineering in Medicine and Biology Society*. He was also the chair of 2016–2018 IEEE Award Committee in Biomedical Engineering and a member of IEEE Medal Panel for Healthcare Technology Award. His research interests include unobtrusive sensing and wearable devices, and neural muscular modeling. He was selected on the lists of China's Most Cited researchers by Elsevier and the top 2% researcher worldwide by Stanford University. He won a number of international awards including IEEEEMBS best journal paper awards, IEEE-EMBS Outstanding Service Award, IEEE-SA 2014 Emerging Technology Award, and Earl Owen Lecture with SMITIBEC2018 in Korea. He is elected as IAMBE fellow, AIMBE fellow and AAIA fellow for his contributions to the development of wearable and m-Health technologies.



Xin Yang received from The Chinese University of Hong Kong, in 2019. He is now an assistant professor with the School of Biomedical Engineering, Shenzhen University, China. His research interests include medical image computing, intelligent ultrasound, cardiac image analysis and computer graphics.



Yining Dong received the BEng degree in electronic engineering from Tsinghua University, Beijing, China, in 2011, and the PhD degree in electrical engineering from the University of Southern California, Los Angeles, California, in 2016. She is currently an assistant professor with the School of Data Science, City University of Hong Kong, Hong Kong. Her research interests include process data analytics, statistical machine learning, smart manufacturing, and new material design.



Zhen Liang (Member, IEEE) received the PhD degree from The Hong Kong Polytechnic University, Hong Kong, in 2013. From 2012 to 2017, she was an algorithm development scientist with NeuroSky, Inc., Hong Kong. From 2018 to 2019, she was a specially-appointed assistant professor of the Graduate School of Informatics, Kyoto University, Japan. She is currently an Assistant Professor with the School of Biomedical Engineering, Medical School, Shenzhen University, China. Her current research interests include affective brain-computer interface, brain encoding and decoding systems, visual attention, and neural engineering.